

Lecture 13, Feb 20-2026

7.3 trigonometric Substitution.

Example. Evaluate $\int \frac{\sqrt{9-x^2}}{x^2} dx$.

Suppose we have a circle or ellipse

$$x^2 + y^2 = r^2$$

or

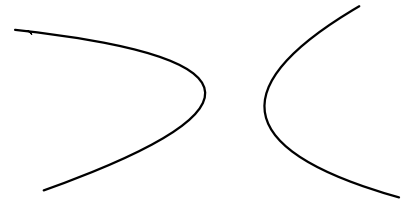
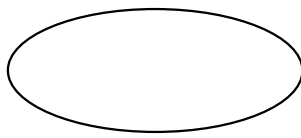
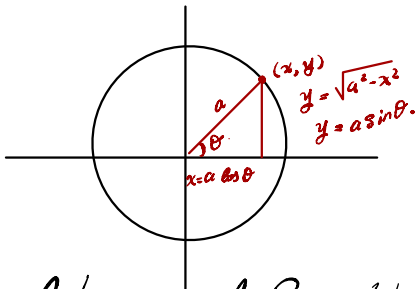
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

$$y = \pm \sqrt{r^2 - x^2}$$

$$y = \pm \sqrt{b^2 - \frac{a^2}{c^2} x^2}$$

$$y = \pm \sqrt{b^2 + \frac{a^2}{c^2} x^2}$$



let simplify the expression $\sqrt{a^2 - x^2}$. let $x = a \cos \theta$. and use the identity $\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow a^2 \cos^2 \theta + a^2 \sin^2 \theta = a^2$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - (a \cos \theta)^2} = \sqrt{a^2 - a^2 \cos^2 \theta} = \sqrt{a^2 \sin^2 \theta}$$

$$= \sqrt{a^2 \underbrace{(1 - \cos^2 \theta)}_{\sin^2 \theta}} = \sqrt{a^2} \sqrt{\sin^2 \theta} = a |\sin \theta|$$

then $\int \underbrace{\sqrt{a^2 - x^2}}_{\text{!!}}$ $dx = \int a \sin \theta \cdot dx = -a^2 \int \sin \theta \cdot \sin \theta d\theta = -a^2 \underbrace{\int \sin^2 \theta d\theta}_{\text{easy!}}$

$x = a \cos \theta \quad dx = -a \sin \theta d\theta$

table of trigonometric Substitutions.

i). $\sqrt{a^2 - x^2}$	$\Rightarrow$	$x = a \sin \theta,$	$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \Rightarrow$	$1 - \sin^2 \theta = \cos^2 \theta.$
ii). $\sqrt{a^2 + x^2}$	$\Rightarrow$	$x = a \tan \theta,$	$-\frac{\pi}{2} < \theta < \frac{\pi}{2} \Rightarrow$	$1 + \tan^2 \theta = \sec^2 \theta.$
iii). $\sqrt{x^2 - a^2}$	$\Rightarrow$	$x = a \sec \theta,$	$0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2} \Rightarrow$	$\sec^2 \theta - 1 = \tan^2 \theta.$

$\int \frac{\sqrt{9-x^2}}{x^2} dx$. Using i). $a=3$ let $x=3\sin\theta$ $dx=3\cos\theta d\theta$

$\sqrt{9-x^2} = \sqrt{9-9\sin^2\theta} = 3|\cos\theta|$ $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ then since $\cos\theta \geq 0$ for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ then $\sqrt{9-x^2} = 3\cos\theta$.

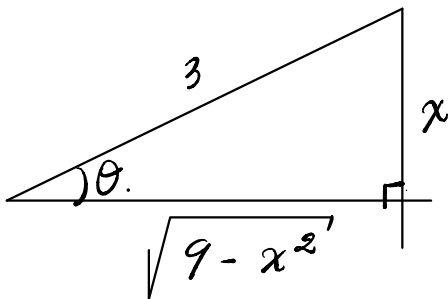
$$\int \frac{\sqrt{9-x^2}}{x^2} dx = \int \frac{3\cos\theta}{9\sin^2\theta} \cdot (3\cos\theta d\theta) = \int \frac{\cos^2\theta}{\sin^2\theta} d\theta$$

$$= \int \cot^2\theta d\theta$$

$$= \int (\csc^2\theta - 1) d\theta$$

$$= \int \csc^2\theta d\theta - \int d\theta$$

$$= -\cot\theta - \theta + C$$



$$x = 3\sin\theta$$

$$\cot\theta = \frac{1}{\tan\theta} = \frac{\text{adj}}{\text{opp}} = \frac{\sqrt{9-x^2}}{x}$$

$$\theta = \arcsin\left(\frac{x}{3}\right)$$

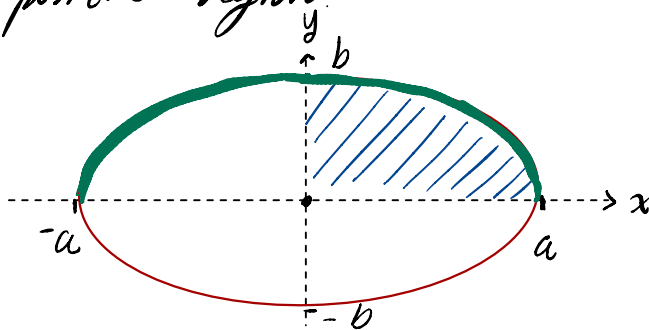
$$= -\frac{1}{x} \sqrt{9-x^2} - \arcsin\left(\frac{x}{3}\right) + C$$

Example. Find the area enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

in the positive region.

Sln.



$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} \Rightarrow y = \sqrt{b^2 - \frac{b^2 x^2}{a^2}}$$

$$y^2 = b^2 - \frac{b^2 x^2}{a^2}$$

$$\begin{aligned}
 A &= \int_0^a \sqrt{b^2 - \frac{b^2 x^2}{a^2}} dx = \int_0^a \sqrt{\frac{b^2(a^2 - x^2)}{a^2}} dx \\
 &= \int_0^a \sqrt{\frac{b^2}{a^2}} \cdot \sqrt{a^2 - x^2} dx \\
 &= \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} dx
 \end{aligned}$$

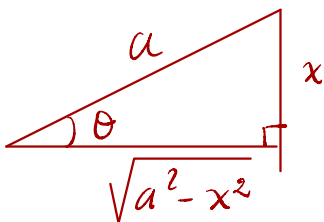
let $x = a \sin \theta$ $dx = a \cos \theta d\theta$

if $x=0$ then $\theta=0$

if $x=a$ then $\sin \theta = 1$ $\theta = \frac{\pi}{2}$

$$\begin{aligned}
 &= \frac{b}{a} \int_0^{\pi/2} \sqrt{a^2(1 - \sin^2 \theta)} a \cos \theta d\theta \\
 &= \frac{b}{a} \int_0^{\pi/2} a^2 \cos^2 \theta d\theta \\
 &= ab \int_0^{\pi/2} \cos^2 \theta d\theta \\
 &= ab \int_0^{\pi/2} \frac{1}{2} (1 + \cos 2\theta) d\theta \\
 &= \frac{ab}{2} \left[\theta \Big|_0^{\pi/2} + \frac{1}{2} \sin 2\theta \Big|_0^{\pi/2} \right] \\
 &= \frac{ab}{2} \left[\frac{\pi}{2} \right] = \frac{\pi}{4} ab.
 \end{aligned}$$

$$\begin{aligned}
 \frac{ab}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^a &= \left[\sin^{-1}\left(\frac{x}{a}\right) + \frac{1}{2} (2 \sin \theta \cos \theta) \right]_0^a \frac{ab}{2} \\
 &= \left[\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{a} \cdot \frac{\sqrt{a^2 - x^2}}{a} \right]_0^a \frac{ab}{2} \\
 &= \left[\sin^{-1}(1) \right] \frac{ab}{2} = \frac{\pi}{4} ab.
 \end{aligned}$$



Example. Find $\int \frac{1}{x^2 \sqrt{x^2+4}} dx$.

sln. let $x = 2 \tan \theta$ $dx = 2 \sec^2 \theta d\theta$.

$$\sqrt{x^2+4} = \sqrt{4 \tan^2 \theta + 4} = 2 \sqrt{\tan^2 \theta + 1} = 2 \sec \theta.$$

$$\int \frac{1}{x^2 \sqrt{x^2+4}} dx = \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta \cdot (2 \sec \theta)} = \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta.$$

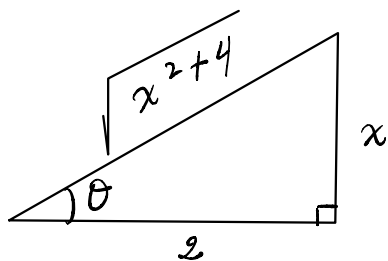
$$= \frac{1}{4} \int \frac{\frac{1}{\cos \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta}} d\theta$$

$$= \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta \quad ; \begin{array}{l} u = \sin \theta \\ du = \cos \theta d\theta \end{array}$$

$$= \frac{1}{4} \int \frac{du}{u^2}$$

$$= -\frac{1}{4} \cdot \frac{1}{u} + C$$

$$= -\frac{1}{4} \cdot \frac{1}{\sin \theta} + C.$$



$$\sin \theta = \frac{x}{\sqrt{x^2+4}}$$

$$= -\frac{1}{4} \cdot \frac{\sqrt{x^2+4}}{x} + C.$$